EE 66 Signals, Dynamics, and Information
Fall 2026 Homework 8

This homework is due Friday, October 23, 2026 at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

On occasion, a problem set contains one or more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

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1. (Optional) Eigenvalues and Determinants: Proofs

Useful facts. The following properties may be helpful:

- If $\mathbf{Av} = \lambda \mathbf{v}$ for some nonzero vector $\mathbf{v}$, then $\mathbf{v}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ.
- The characteristic polynomial of a square matrix $\mathbf{A}$ is $\det(\lambda \mathbf{I} - \mathbf{A})$.
- The eigenvalues of $\mathbf{A}$ are the roots of its characteristic polynomial.
- The determinant is defined only for square matrices.
- $\det(\mathbf{A}) = \det(\mathbf{A}^\top)$.
- $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$.
- $\det(\mathbf{A}) = 0$ if and only if $\mathbf{A}$ is not invertible.

Now, onto proving!

(a) Let $\mathbf{A}$ be an invertible matrix. Show that if λ is an eigenvalue of $\mathbf{A}$, then $\dfrac{1}{\lambda}$ is an eigenvalue of $\mathbf{A}^{-1}$.

Solution: $\mathbf{A}$ is invertible,
$\Rightarrow \mathbf{Ax} = \mathbf{b}$ has a unique solution for all $\mathbf{b}$.
$\Rightarrow \mathbf{Ax} = \mathbf{0}$ has a unique solution.
$\Rightarrow \mathbf{x} = \mathbf{0}$ is the only solution to $\mathbf{Ax} = \mathbf{0}$
$\Rightarrow \mathbf{Ax} = \mathbf{0}$ has no non-zero vectors $\mathbf{x}$ that satisfy it.

Therefore, 0 is not an eigenvalue. Let $\mathbf{v}$ be the eigenvector of $\mathbf{A}$ corresponding to λ.

$$\mathbf{Av} = \lambda \mathbf{v}$$

Since we know that $\mathbf{A}$ is invertible, we can left-multiply both sides by $\mathbf{A}^{-1}$.

$$\mathbf{A}^{-1}\mathbf{Av} = \lambda \mathbf{A}^{-1}\mathbf{v}$$

$$\mathbf{v} = \lambda \mathbf{A}^{-1}\mathbf{v}$$

$$\mathbf{A}^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$$

Note that it is important for our eigenvalues to be non-zero, since otherwise $\frac{1}{\lambda} = \frac{1}{0}$ is undefined.

(b) Given that A is a square matrix, show that $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.

Solution: The eigenvalues of $\mathbf{A}$ satisfy $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$. However, because determinants of a matrix and its transpose are equal, then it is the case that

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \det\left((\lambda \mathbf{I} - \mathbf{A})^\top\right) = \det\left(\lambda \mathbf{I}^\top - \mathbf{A}^\top\right) = \det(\lambda \mathbf{I} - \mathbf{A}^\top).$$

For the choice of λ that makes $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$, we have that $\det(\lambda \mathbf{I} - \mathbf{A}^\top) = 0$ as well. So the eigenvalues of $\mathbf{A}$ are the same as those of $\mathbf{A}^\top$.

(c) Show that the nonzero eigenvalues of $\mathbf{A}^\top\mathbf{A}$ are the same as the nonzero eigenvalues of $\mathbf{AA}^\top$.

Solution: The eigenvalues of $\mathbf{A}^\top\mathbf{A}$ satisfy $\mathbf{A}^\top\mathbf{Av} = \lambda \mathbf{v}$, for some vector $\mathbf{v}$. Left-multiply both sides of the equality by $\mathbf{A}$:

$$\mathbf{A}(\mathbf{A}^\top\mathbf{Av}) = \mathbf{A}(\lambda \mathbf{v})$$

$$(\mathbf{A}\mathbf{A}^\top)(\mathbf{A}\mathbf{v}) = \lambda(\mathbf{A}\mathbf{v})$$

The last step follows from the associativity of matrix multiplication, and shows us that if $\mathbf{A}^\top\mathbf{A}$ has nonzero eigenvalue λ, then $(\mathbf{A}\mathbf{A}^\top)$ has an eigenvector $\mathbf{A}\mathbf{v}$ associated with the same eigenvalue λ. Thus, the nonzero eigenvalues of $\mathbf{A}^\top\mathbf{A}$ are the same as the nonzero eigenvalues of $\mathbf{A}\mathbf{A}^\top$.

(d) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ and define $\mathbf{B} = b\mathbf{A} + c\mathbf{I}$ for scalars $b, c \in \mathbb{R}$.

i. Suppose λ is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v}$. Show that $\mathbf{v}$ is also an eigenvector of $\mathbf{B}$ and find its corresponding eigenvalue μ.
Solution: Since $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$,

$$\mathbf{B}\mathbf{v} = (b\mathbf{A} + c\mathbf{I})\mathbf{v} = b\lambda\mathbf{v} + c\mathbf{v} = (b\lambda + c)\mathbf{v}.$$

So $\mathbf{v}$ is an eigenvector of $\mathbf{B}$ with corresponding eigenvalue $\mu = b\lambda + c$.

ii. Suppose the eigenvalues of $\mathbf{A}$ are $\lambda_1, \ldots, \lambda_n$. Use part (a) to show that the corresponding eigenvalues of $\mathbf{B}$ are

$$\mu_i = b\lambda_i + c, \qquad i = 1, \ldots, n$$

Solution: For each eigenvalue λ_i of $\mathbf{A}$, choose the corresponding eigenvector $\mathbf{v}_i$. By part (a),

$$\mathbf{B}\mathbf{v}_i = (b\lambda_i + c)\mathbf{v}_i$$

Therefore, each eigenvalue λ_i of $\mathbf{A}$ is scaled by b and shifted by c to give the corresponding eigenvalue $\mu_i = b\lambda_i + c$ of $\mathbf{B}$.

(e) Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$. Assume that $\mathbf{A}$ is invertible, but $\mathbf{B}$ is not invertible. Show that neither $\mathbf{AB}$ nor $\mathbf{BA}$ is invertible.

Solution: There are two ways to show this:

i. The easiest way to prove this is to use determinants. Since $\mathbf{B}$ is not invertible, $\det(\mathbf{B}) = 0$.

$$\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}) = \det(\mathbf{A}) \cdot 0 = 0$$

$$\det(\mathbf{BA}) = \det(\mathbf{B})\det(\mathbf{A}) = 0 \cdot \det(\mathbf{A}) = 0$$

Since $\det(\mathbf{AB}) = \det(\mathbf{BA}) = 0$, neither $\mathbf{AB}$ nor $\mathbf{BA}$ is invertible.

ii. Since $\mathbf{B}$ is not invertible, there must exist a non-zero vector $\mathbf{x}$, such that $\mathbf{B}\mathbf{x} = \mathbf{0}$. We then left-multiply both sides by $\mathbf{A}$.

$$\mathbf{A}(\mathbf{B}\mathbf{x}) = \mathbf{A}\mathbf{0} \implies \mathbf{A}\mathbf{B}\mathbf{x} = \mathbf{0}$$

Since $\mathbf{x}$ is a non-zero vector, $\mathbf{AB}$ has a non-trivial null space, so it is not invertible.
To prove that $\mathbf{BA}$ is not invertible, we know that there exists a vector $\mathbf{y}$, such that $\mathbf{A}^{-1}\mathbf{x} = \mathbf{y} \iff \mathbf{x} = \mathbf{A}\mathbf{y}$ since the inverse $\mathbf{A}^{-1}$ exists. $\mathbf{y}$ is a non-zero vector because $\mathbf{x}$ is a non-zero vector. But for some $\mathbf{x}$ in the null space of $\mathbf{B}$,

$$\mathbf{B}\mathbf{x} = \mathbf{0} \implies \mathbf{B}(\mathbf{A}\mathbf{y}) = \mathbf{B}\mathbf{A}\mathbf{y} = \mathbf{0}$$

Since $\mathbf{y}$ is a non-zero vector, $\mathbf{BA}$ has a non-trivial null space, so it is not invertible as well.

2. (Optional) Reverse Mechanical Diagonalization

All calculations in this problem are intended to be done by hand, but you can use a computer to check your work.

(a) Diagonalize the matrix $\mathbf{A}$ below. That is, find $\mathbf{V}$, $\mathbf{\Lambda}$, and $\mathbf{V}^{-1}$ such that $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$, where $\mathbf{\Lambda}$ is diagonal with the eigenvalues of $\mathbf{A}$ along its diagonal.

$$\mathbf{A} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} \tag{1}$$

Solution: We first find the eigenvalues from the characteristic polynomial:

$$\det(\lambda\mathbf{I} - \mathbf{A}) = \det\left(\begin{bmatrix} \lambda - \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \lambda - \frac{3}{2} \end{bmatrix}\right) = \left(\lambda - \frac{3}{2}\right)^2 - \frac{1}{4} = \lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2) = 0, \tag{2}$$

so $\mathbf{A}$ has eigenvalues $\lambda = 1$ and $\lambda = 2$. As a check, the trace of $\mathbf{A}$ is $\frac{3}{2} + \frac{3}{2} = 3 = 1 + 2$, and $\det(\mathbf{A}) = \frac{9}{4} - \frac{1}{4} = 2 = 1 \cdot 2$.
Now we find an eigenvector for each eigenvalue. For $\lambda = 1$,

$$\text{Null}\,(\mathbf{I} - \mathbf{A}) = \text{Null}\begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \text{span}\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right), \tag{3}$$

and for $\lambda = 2$,

$$\text{Null}\,(2\mathbf{I} - \mathbf{A}) = \text{Null}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right). \tag{4}$$

Putting the eigenvectors in the columns of $\mathbf{V}$, in the same order as the eigenvalues along $\mathbf{\Lambda}$,

$$\mathbf{V} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{\Lambda} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}. \tag{5}$$

Since $\det(\mathbf{V}) = 2$, the inverse of this 2×2 matrix is

$$\mathbf{V}^{-1} = \frac{1}{\det(\mathbf{V})}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}. \tag{6}$$

Multiplying out to check,

$$\mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix} = \mathbf{A}. \tag{7}$$

(b) Now we run the process backwards. Suppose you are told only that a 2×2 matrix $\mathbf{B}$ has

$$\lambda_1 = 2 \text{ with eigenvector } \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \lambda_2 = -1 \text{ with eigenvector } \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \tag{8}$$

Find $\mathbf{B}$. Then verify your answer by checking that $\mathbf{B}\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ and $\mathbf{B}\mathbf{v}_2 = \lambda_2\mathbf{v}_2$.

Solution: In part (a) we started from $\mathbf{A}$ and worked out $\mathbf{V}$ and $\mathbf{\Lambda}$. Here we are handed those two pieces and need to assemble the matrix, so we use the same relationship in the other direction: $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$, where the columns of $\mathbf{V}$ are the given eigenvectors and the diagonal of $\mathbf{\Lambda}$ holds the matching eigenvalues, in the same order:

$$\mathbf{V} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{\Lambda} = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}. \tag{9}$$

Again $\det(\mathbf{V}) = (1)(2) - (1)(1) = 1$, so

$$\mathbf{V}^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \tag{10}$$

and therefore

$$\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ 6 & -4 \end{bmatrix}. \tag{11}$$

Checking against the data we were given,

$$\mathbf{B}\mathbf{v}_1 = \begin{bmatrix} 5 & -3 \\ 6 & -4 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2\mathbf{v}_1, \qquad \mathbf{B}\mathbf{v}_2 = \begin{bmatrix} 5 & -3 \\ 6 & -4 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} = -\mathbf{v}_2. \tag{12}$$

It is also worth checking the trace and determinant, which must agree with the eigenvalues: $5 + (-4) = 1 = 2 + (-1)$ and $(5)(-4) - (-3)(6) = -2 = (2)(-1)$.

(c) Your answer to part (b) came from one particular choice of $\mathbf{V}$ and $\mathbf{\Lambda}$, but would a different choice of eigenvectors change it? For each of the following, state whether the matrix $\mathbf{B}$ can still be built, and if so, whether it changes your answer to part (b). Explain your reasoning.

(i) You use $5\mathbf{v}_1$ in place of $\mathbf{v}_1$.

(ii) You swap both the columns of $\mathbf{V}$, and the diagonal entries of $\mathbf{\Lambda}$.

(iii) You are told instead that the second eigenvector is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, with $\lambda_2 = -1$ as before.

Solution:

(i) No Change. We can represent the scaling as $\mathbf{D} = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$:

$$\mathbf{V}\mathbf{D} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 5 & 2 \end{bmatrix}, \qquad (\mathbf{V}\mathbf{D})^{-1} = \frac{1}{\det(\mathbf{V}\mathbf{D})}\begin{bmatrix} 2 & -1 \\ -5 & 5 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 2 & -1 \\ -5 & 5 \end{bmatrix} = \mathbf{D}^{-1}\mathbf{V}^{-1}, \tag{13}$$

where $\det(\mathbf{V}\mathbf{D}) = \det(\mathbf{V})\det(\mathbf{D}) = 5$, so the factor the scaling introduces into the columns is exactly cancelled by the reciprocal factor in the inverse. Since $\mathbf{D}$ and $\mathbf{\Lambda}$ are diagonal they commute, so

$$(\mathbf{V}\mathbf{D})\mathbf{\Lambda}(\mathbf{V}\mathbf{D})^{-1} = \mathbf{V}\mathbf{D}\mathbf{\Lambda}\mathbf{D}^{-1}\mathbf{V}^{-1} = \mathbf{V}\mathbf{\Lambda}\mathbf{D}\mathbf{D}^{-1}\mathbf{V}^{-1} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1} = \mathbf{B}, \tag{14}$$

We can also verify by directly multiplying:

$$\begin{bmatrix} 5 & 1 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} \frac{2}{5} & -\frac{1}{5} \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ 6 & -4 \end{bmatrix} = \mathbf{B}. \tag{15}$$

Only the *direction* of an eigenvector matters! The length doesn't matter, since any scaling to eigenvectors cancels itself out.

(ii) No Change. As long as the columns of $\mathbf{V}$ and the diagonal entries of $\mathbf{\Lambda}$ are reordered together so that each eigenvector stays matched with its own eigenvalue. Changing the order is just relabelling which eigenpair we call "first," and $\mathbf{B}$ does not depend on that labelling.

What *would* change the answer is pairing an eigenvector with the wrong eigenvalue. If you reorder the diagonal of $\mathbf{\Lambda}$ but leave the columns of $\mathbf{V}$ alone, you are describing a different matrix entirely...

(iii) Changes! With a different eigenvector, it now describes a different transformation! $\det(\mathbf{V}) = -2 \neq 0$, so the construction still runs, but $\mathbf{V\Lambda V}^{-1}$ won't return $\mathbf{B}$:

$$\mathbf{V\Lambda V}^{-1} = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{3}{2} \\ \frac{3}{2} & \frac{1}{2} \end{bmatrix} \neq \mathbf{B}. \tag{16}$$

So you can get the same result by scaling eigenvectors or reordering eigenpairs (as long as you keep the eigenvector directions). Pretty neat!

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3. Spectral Theorem for Real Symmetric Matrices

(a) Show that all eigenvalues λ of a real symmetric matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ are real (i.e., $\lambda = \lambda^* \in \mathbb{R}$).

Hint: Let $\mathbf{Av} = \lambda\mathbf{v}$ for a nonzero vector $\mathbf{v} \in \mathbb{C}^n$. Try evaluating the scalar quantity $\mathbf{v}^\top\mathbf{Av}^*$ in two different ways by associating $\mathbf{A}$ with $\mathbf{v}$ on the left versus $\mathbf{v}^*$ on the right.

Solution: Taking the complex conjugate of both sides of the eigenvalue relation $\mathbf{Av} = \lambda\mathbf{v}$, and noting that $\mathbf{A}$ is a real matrix ($\mathbf{A}^* = \mathbf{A}^\top = \mathbf{A}$), we have:

$$\mathbf{Av}^* = \lambda^*\mathbf{v}^*$$

Taking the transpose of $\mathbf{Av} = \lambda\mathbf{v}$ and using symmetry ($\mathbf{A}^\top = \mathbf{A}$), we get:

$$\mathbf{v}^\top\mathbf{A}^\top = \lambda\mathbf{v}^\top \implies \mathbf{v}^\top\mathbf{A} = \lambda\mathbf{v}^\top$$

Now, right-multiply this expression by $\mathbf{v}^*$:

$$\mathbf{v}^\top\mathbf{Av}^* = \lambda\mathbf{v}^\top\mathbf{v}^*$$

Substituting $\mathbf{Av}^* = \lambda^*\mathbf{v}^*$ into the left-hand side:

$$\mathbf{v}^\top(\lambda^*\mathbf{v}^*) = \lambda\mathbf{v}^\top\mathbf{v}^* \implies \lambda^*(\mathbf{v}^\top\mathbf{v}^*) = \lambda(\mathbf{v}^\top\mathbf{v}^*)$$

$$(\lambda - \lambda^*)(\mathbf{v}^\top\mathbf{v}^*) = 0$$

Since $\mathbf{v}$ is an eigenvector, $\mathbf{v} \neq \mathbf{0}$, so $\mathbf{v}^\top\mathbf{v}^* = \|\mathbf{v}\|^2 > 0$. Therefore, we must have $\lambda - \lambda^* = 0$, which implies $\lambda = \lambda^* \in \mathbb{R}$. Since $\lambda \in \mathbb{R}$, its associated eigenvector $\mathbf{v}$ can also be chosen to be real-valued ($\mathbf{v} \in \mathbb{R}^n$).

(b) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a real symmetric matrix. Suppose $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^n$ are eigenvectors of $\mathbf{A}$ corresponding to two distinct real eigenvalues $\lambda_1 \neq \lambda_2$. Show that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal ($\mathbf{v}_2^\top\mathbf{v}_1 = 0$), and explain why this implies that eigenvectors corresponding to distinct eigenvalues are linearly independent.

Hint: Consider two distinct eigenpairs $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$. Try evaluating the expression $\mathbf{v}_2^\top\mathbf{Av}_1$ in two different ways using matrix symmetry $\mathbf{A} = \mathbf{A}^\top$.

Solution: We are given $\mathbf{Av}_1 = \lambda_1\mathbf{v}_1$ and $\mathbf{Av}_2 = \lambda_2\mathbf{v}_2$. Left-multiplying $\mathbf{Av}_1 = \lambda_1\mathbf{v}_1$ by $\mathbf{v}_2^\top$ yields:

$$\mathbf{v}_2^\top\mathbf{Av}_1 = \lambda_1\mathbf{v}_2^\top\mathbf{v}_1$$

Using matrix transposition and symmetry ($\mathbf{A}^\top = \mathbf{A}$), we rewrite the left side:

$$\mathbf{v}_2^\top\mathbf{Av}_1 = \mathbf{v}_2^\top\mathbf{A}^\top\mathbf{v}_1 = (\mathbf{Av}_2)^\top\mathbf{v}_1$$

Since $\mathbf{Av}_2 = \lambda_2\mathbf{v}_2$, this becomes:

$$(\lambda_2\mathbf{v}_2)^\top\mathbf{v}_1 = \lambda_2\mathbf{v}_2^\top\mathbf{v}_1$$

Equating the two expressions for $\mathbf{v}_2^\top\mathbf{Av}_1$:

$$\lambda_2\mathbf{v}_2^\top\mathbf{v}_1 = \lambda_1\mathbf{v}_2^\top\mathbf{v}_1 \implies (\lambda_2 - \lambda_1)\mathbf{v}_2^\top\mathbf{v}_1 = 0$$

Because the eigenvalues are distinct ($\lambda_1 \neq \lambda_2$), $\lambda_2 - \lambda_1 \neq 0$. Thus, we must have $\mathbf{v}_2^\top\mathbf{v}_1 = 0$, proving that $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal.

Since mutually orthogonal non-zero vectors are always linearly independent, any set of eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix is linearly independent.

(c) Suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ is symmetric and has n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ with corresponding normalized eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ (such that $\|\mathbf{v}_i\| = 1$). Use your results from parts (a) and (b) to show that $\mathbf{A}$ can be written in the form $\mathbf{A} = \mathbf{V}\Lambda\mathbf{V}^\top$, where $\mathbf{V}$ is an orthogonal matrix and Λ is a diagonal matrix containing the eigenvalues.

Solution: By part (a), all eigenvalues $\lambda_1, \ldots, \lambda_n$ are real. By part (b), eigenvectors corresponding to distinct eigenvalues are mutually orthogonal. Since each vector is normalized ($\|\mathbf{v}_i\| = 1$), the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ forms an orthonormal basis for $\mathbb{R}^n$.

Constructing $\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}$, the orthonormality condition $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$ implies:

$$\mathbf{V}^\top \mathbf{V} = \mathbf{I} \implies \mathbf{V}^{-1} = \mathbf{V}^\top$$

From standard matrix diagonalization, $\mathbf{AV} = \mathbf{V}\Lambda$, where $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$. Right-multiplying by $\mathbf{V}^{-1}$:

$$\mathbf{A} = \mathbf{V}\Lambda\mathbf{V}^{-1}$$

Substituting $\mathbf{V}^{-1} = \mathbf{V}^\top$ yields the Spectral Theorem decomposition:

$$\mathbf{A} = \mathbf{V}\Lambda\mathbf{V}^\top$$

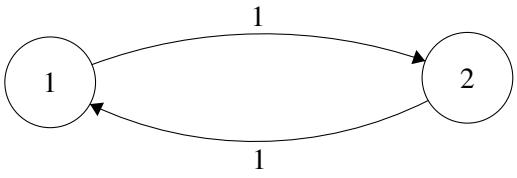
4. PageRank

In this problem, we consider using the adjacency matrices to represent "state transitions" in the setting of web traffic, and how users flow between different websites.

The “connectivity matrix”, **C**, can be constructed using the state transition diagram as follows: entries c_{ji} represent the *proportion* of the people who are at website i that click the link for website j. This is slightly different from the adjacency matrix. Its values, c_{ji}, are the *proportion* of the people who are at website i that click the link for website j. It is associated with a graph that has labels stating the percentage (in our case of people) that is moving from one state to another.

Once computed, an eigenvector with eigenvalue 1 will have values which correspond to the steady-state frequency (i.e. the distribution in the limit as time goes to infinity) for the fraction of people for each webpage. When the elements of this eigenvector are made to **sum to one** (to conserve population), the ith element of the eigenvector will correspond to the fraction of people on the ith website.

(a) For graph A shown below, what are the steady-state frequencies (i.e. fraction of visitors in steady-state) for the two webpages? Graph A has weights in place to help you construct the connectivity matrix. Remember to ensure that your steady state-frequencies sum to 1 to maintain conservation.



Graph A

Solution:

To determine the steady-state frequencies for the two pages, we need to find the appropriate eigenvector of the connectivity matrix. In this case, we are trying to determine the proportion of people who would be on a given page at steady state.

The connectivity matrix of graph A:

$$\mathbf{C} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{17}$$

To determine the eigenvalues of this matrix:

$$\det\left(\begin{bmatrix} \lambda & -1 \\ -1 & \lambda \end{bmatrix}\right) = \lambda^2 - 1 = 0 \tag{18}$$

So we get $\lambda = 1, -1$.

The steady state vector is the eigenvector that corresponds to $\lambda = 1$. To find the eigenvector,

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \tag{19}$$

The sum of the values of the vector should equal 1 since the number of people is conserved, so our conditions are:

$$v_1 + v_2 = 1$$
$$v_1 = v_2$$

The steady-state frequency eigenvector is $\begin{bmatrix}0.5\\0.5\end{bmatrix}$ and each webpage has a steady-state frequency of 0.5.

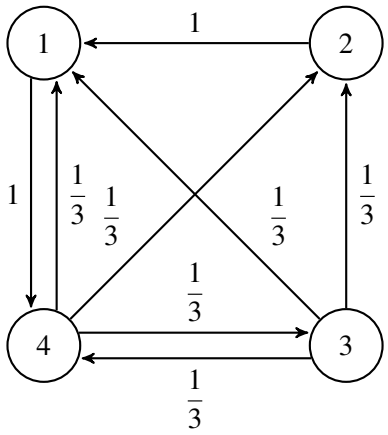
(b) In Graph A, notice that the system is periodic: a visitor at Page 1 at time n will always be at Page 2 at time $n+1$. This corresponds to the eigenvalue $\lambda = -1$ (an Nth root of unity where $N = 2$).
Using this observation, find the PageRank score vector, s^*.
Hint: Think about the mean of the oscillations over many steps.
Solution: For a network with N-periodic dynamics, the PageRank score vector s^* is defined as the mean over one full period (s_{AVG}). If we assume an initial distribution $s(0) = \begin{bmatrix}a\\1-a\end{bmatrix}$, the state at the next time step is $s(1) = \begin{bmatrix}1-a\\a\end{bmatrix}$. The mean of this sequence is:

$$s_{AVG} = \frac{1}{2}\sum_{n=0}^{1} s[n] = \frac{1}{2}\left(\begin{bmatrix}a\\1-a\end{bmatrix} + \begin{bmatrix}1-a\\a\end{bmatrix}\right) = \begin{bmatrix}\frac{a+(1-a)}{2}\\\frac{(1-a)+a}{2}\end{bmatrix} = \begin{bmatrix}0.5\\0.5\end{bmatrix} \tag{20}$$

Therefore, even though the population is constantly moving, the average proportion of time spent on each website (the PageRank score) is 0.5 for both Page 1 and Page 2. This matches the "importance" we would intuitively assign to two identically linked pages.

(c) For graph B, what are the steady-state frequencies for the webpages? You may use IPython and the Numpy command `numpy.linalg.eig` for this. Graph B is shown below, with weights in place to help you construct the connectivity matrix.
Hint: `numpy.linalg.eig` *returns eigenvectors and eigenvalues. The eigenvectors are arranged in a matrix in column-major order. In other words, given eigenvectors*
$\mathbf{v_1} = \begin{bmatrix}v_{11}\\v_{12}\end{bmatrix}$ and $\mathbf{v_2} = \begin{bmatrix}v_{21}\\v_{22}\end{bmatrix}$
NumPy will return:

$$\begin{bmatrix}v_{11} & v_{21}\\v_{12} & v_{22}\end{bmatrix} \tag{21}$$



Graph B

Solution:

To determine the steady-state frequencies, we need to create the connectivity matrix $\mathbf{C}$ first.

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ 1 & 0 & \frac{1}{3} & 0 \end{bmatrix}$$

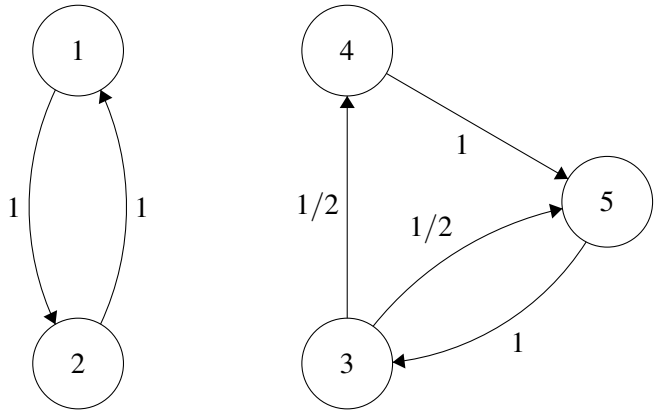
One possible eigenvector associated with eigenvalue 1 is $\begin{bmatrix} -0.61 & -0.31 & -0.23 & -0.69 \end{bmatrix}^\top$ (found using IPython). Scaling it by

$$\frac{1}{(-0.61+(-0.31)+(-0.23)+(-0.69))}$$

so the elements sum to 1, we get $\begin{bmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{8} & \frac{3}{8} \end{bmatrix}^\top$

These are the steady-state frequencies for the pages.

(d) Graph D with weights in place is shown below. Find the eigenspace that corresponds to the steady-state for graph D. How many independent systems (disjoint sets of webpages) are there in graph D versus in graph B? What is the dimension of the eigenspace corresponding to the steady-state for graph D? You may use IPython to compute the eigenvalues and eigenvectors again.



Graph D

Solution:

The connectivity matrix for graph D is

$$\mathbf{C} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 1 & 0 \end{bmatrix}$$

Using IPython, we find that the eigenspace associated with $\lambda = 1$ is spanned by the vectors

$$\begin{bmatrix} 0 \\ 0 \\ 0.4 \\ 0.2 \\ 0.4 \end{bmatrix}, \begin{bmatrix} 0.5 \\ 0.5 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

While any linear combination of these vectors is an eigenvector, these two particular vectors have a nice interpretation.

The first eigenvector describes the steady-state frequencies for the last three webpages, and the second vector describes the steady-state frequencies for the first two webpages. This makes sense since there are essentially "two internets", or two disjoint sets of webpages. Surfers cannot transition between the two, so you cannot assign steady-state frequencies to webpage 1 and webpage 2 relative to the rest. This is why the eigenspace corresponding to the steady-state has dimension 2.

> Assuming that each set of steady-state frequencies needs to add to 1, the first assigns steady-state frequencies of 0.4, 0.2, 0.4 to webpage 3, webpage 4, and webpage 5, respectively. The second assigns steady-state frequencies of 0.5 to both webpage 1 and webpage 2.

5. Perpetual Motion Machine

For your science fair project, you are designing a **perpetual motion machine**, a machine designed to sustain its own motion for eternity, without external energy! You think you've come across the perfect design, but once you build it, you realize it isn't working as expected.

Your design involves the motion of a ball throughout a loosely defined path, which you believe should never slow down by **conservation of mechanical energy**. As a result, you thought, the ball should continue cycling through the path. However, after measuring its speed at a specific point on the path across multiple cycles, you realize that the ball is actually slowing down!

The ball's velocity, at the point of measurement, after the nth cycle is denoted by the vector $\mathbf{v}[n] = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$. In this question, a cycle refers to a single pass of the ball through the path, passing its initial starting point.

Note: Feedback control is beyond the scope of this course. *For this question, you only need to understand state space with inputs. If you're curious, you can learn more about feedback control in EE C128 or EECS C106A/B.*

(a) The initial velocity of the ball at the point of measurement is given by the vector $\mathbf{v}[0] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. You take a few more measurements, and find the velocities after the first and second cycles are as follows:

$$\mathbf{v}[1] = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}, \mathbf{v}[2] = \begin{bmatrix} 0.25 \\ 0.6 \end{bmatrix}$$

Is the ball slowing down?

Hint: The magnitude of the velocity vector lets you know its speed.

Solution:

$$\|\mathbf{v}[0]\| = 1, \|\mathbf{v}[1]\| = \frac{\sqrt{2}}{2} \approx 0.7, \|\mathbf{v}[2]\| = \sqrt{0.25^2 + 0.6^2} = 0.65$$

We can see that the ball's speed is decreasing.

(b) We can represent the relationship between the measurements from (a) as a difference equation of the form $\mathbf{v}[n+1] = \mathbf{A}\mathbf{v}[n]$. For the measurements shown above, **solve for A**.

Solution: Since our vectors are consistently 2×1, we can see that $\mathbf{A}$ must be 2×2. We can begin by representing it as four variables:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We have two equations, one relating $\mathbf{v}[0]$ to $\mathbf{v}[1]$, and another relating $\mathbf{v}[1]$ to $\mathbf{v}[2]$:

$$\mathbf{v}[1] = \mathbf{A}\mathbf{v}[0] \implies \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\mathbf{v}[2] = \mathbf{A}\mathbf{v}[1] \implies \begin{bmatrix} 0.25 \\ 0.6 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$

We can convert these into two sets of linear equations. For the first relation:

$$1a + 0b = 0.5$$

$$1c + 0d = 0.5$$

For the second relation:

$$0.5a + 0.5b = 0.25$$

$$0.5c + 0.6d = 0.6$$

We can see the first set is already in solved form:

$$1a + 0b = 0.5 \implies a = 0.5$$

$$1c + 0d = 0.5 \implies c = 0.5$$

Substituting a and c into our second set of equations gives us:

$$0.5(0.5) + 0.5b = 0.25 \implies 0.25 + 0.5b = 0.25 \implies b = 0$$

$$0.5(0.5) + 0.5d = 0.6 \implies 0.25 + 0.5d = 0.6 \implies d = 0.7$$

Thus, our matrix $\mathbf{A}$ is:

$$\mathbf{A} = \begin{bmatrix} 0.5 & 0 \\ 0.5 & 0.7 \end{bmatrix}$$

(c) As you keep measuring the ball's velocity, you wonder whether it could ever settle at a non-zero velocity. Since you have a rough model of the system, you decide to analyze it mathematically.

Find the eigenvalues of A. Using this, demonstrate that, regardless of its initial velocity, the ball will always come to a complete stop over time.

Hint: Write $\mathbf{v}[n] = \mathbf{A}^n\mathbf{v}[0]$ *and use* $\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$ *to examine the limit as* $n \to \infty$.

Solution: Since $\mathbf{A}$ is lower triangular, we can read the eigenvalues off the diagonal:

$$\lambda_1 = 0.5, \lambda_2 = 0.7$$

Since we have two distinct, real eigenvalues, we must have two linearly independent eigenvectors.

Given an arbitrary initial velocity $\mathbf{v}[0]$, we can denote the velocity $\mathbf{v}[n]$ as follows:

$$\mathbf{v}[n] = \mathbf{A}^n\mathbf{v}[0]$$

$$= \mathbf{V}\mathbf{\Lambda}^n\mathbf{V}^{-1}\mathbf{v}[0] = \mathbf{V}\begin{bmatrix} 0.7 & 0 \\ 0 & 0.5 \end{bmatrix}^n\mathbf{V}^{-1}\mathbf{v}[0] = \mathbf{V}\begin{bmatrix} 0.7^n & 0 \\ 0 & 0.5^n \end{bmatrix}\mathbf{V}^{-1}\mathbf{v}[0]$$

As $n \to \infty$, we can see that our eigenvalue matrix $\mathbf{\Lambda}$ decays to the $\mathbf{0}$ matrix.

Thus, as $n \to \infty$, we have:

$$\lim_{n\to\infty} \mathbf{v}[n] = \mathbf{V}\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\mathbf{V}^{-1}\mathbf{v}[0] = \mathbf{0}$$

Since $\mathbf{v}[n] \to \mathbf{0}$ as $n \to \infty$, the ball cannot settle at a non-zero velocity; it always comes to a complete stop.

(d) Since the ball always slows to a stop on its own, keeping your "perpetual" motion machine alive requires external energy. You add a motor that boosts the ball's velocity at the point of measurement.

The motor power during the measurement after cycle n is denoted by the vector $\mathbf{u}[n]$, representing the effective power of the motor in the x and y directions.

The motor's effect on the velocity of the ball is a linear transformation of $\mathbf{u}[n]$, denoted by $\mathbf{Bu}[n]$, where $\mathbf{B}$ is a matrix. Now, our vector difference equation is as follows:

$$\mathbf{v}[n+1] = \mathbf{Av}[n] + \mathbf{Bu}[n]$$

In an attempt to find a suitable input such that the speed of the ball remains constant, you initially try to use a constant motor power input $\mathbf{u}[n] = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$ for all n and $\mathbf{B} = \mathbf{I}$.

Compute the value of $\mathbf{v}[1]$, assuming $\mathbf{v}[0] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Is this a suitable control input for your science fair project?

Solution:

$$\mathbf{v}[1] = \mathbf{Av}[0] + \mathbf{Bu}[0] = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 5.5 \end{bmatrix}$$

Evidently, the ball is speeding up significantly. Thus, this input is not suitable for your science fair project.

(e) Now, you get creative and try to adapt the motor input to the velocity of the ball. Assume we use the motor power input $\mathbf{u}[n] = \mathbf{v}[n]$ for all n and $\mathbf{B} = \mathbf{I}$. **Find $\|\mathbf{v}[n]\|$ as $n \to \infty$. Is this a suitable control input for your science fair project?**

Solution: Using our defined control scheme, we have:

$$\mathbf{v}[n+1] = \mathbf{Av}[n] + \mathbf{Bu}[n] = \mathbf{Av}[n] + \mathbf{Bv}[n] = (\mathbf{A} + \mathbf{B})\mathbf{v}[n]$$

Since $\mathbf{A} = \begin{bmatrix} 0.5 & 0 \\ 0.5 & 0.7 \end{bmatrix}$, and $\mathbf{B} = \mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, we have:

$$\mathbf{v}[n+1] = \left(\begin{bmatrix} 0.5 & 0 \\ 0.5 & 0.7 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \mathbf{v}[n] = \begin{bmatrix} 1.5 & 0 \\ 0.5 & 1.7 \end{bmatrix} \mathbf{v}[n]$$

Using the same argument as part (b), we can see that:

$$\mathbf{v}[n] = \mathbf{V} \begin{bmatrix} 1.7^n & 0 \\ 0 & 1.5^n \end{bmatrix} \mathbf{V}^{-1} \mathbf{v}[0]$$

As $n \to \infty$, we can see that $1.7^n \to \infty$, and $1.5^n \to \infty$. Thus, $\|\mathbf{v}[n]\| \to \infty$. This is not a suitable solution for your science project, as the ball will speed up until the motor hits its power limits, and you want to demonstrate that it will maintain a constant speed equivalent to its starting speed.

(f) Now, you use a tuned **feedback control** input. Instead of a fixed boost, let the motor react to the ball: at each cycle it reads the ball's current velocity and sets its power from that reading, so the input is

$$\mathbf{u}[n] = \mathbf{Fv}[n] \text{ for all } n, \text{ where } \mathbf{F} = \begin{bmatrix} 0.5 & 0 \\ -0.5 & 0.3 \end{bmatrix}$$

Choosing the input from a measurement of the system is what makes this **feedback control**: the ball's velocity is fed back to the motor. These sorts of systems are a really interesting topic, and can be used in a wide variety of applications; in later classes, you'll learn to construct these inputs yourself!

Once again, the overall vector difference equation is still as follows:

$$\mathbf{v}[n+1] = \mathbf{A}\mathbf{v}[n] + \mathbf{B}\mathbf{u}[n]$$

with $\mathbf{B} = \mathbf{I}$.

For this question, **show that, using the given control system, the measured velocity of the ball stays constant.**

Solution: Using our defined control system, we have:

$$\mathbf{v}[n+1] = \mathbf{A}\mathbf{v}[n] + \mathbf{B}\mathbf{u}[n] = \mathbf{A}\mathbf{v}[n] + \mathbf{B}\mathbf{F}\mathbf{v}[n] = (\mathbf{A} + \mathbf{B}\mathbf{F})\mathbf{v}[n]$$

We can see that $(\mathbf{A} + \mathbf{B}\mathbf{F})$ is defined as follows:

$$\mathbf{A} + \mathbf{B}\mathbf{F} = \begin{bmatrix} 0.5 & 0 \\ 0.5 & 0.7 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 \\ -0.5 & 0.3 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0 \\ 0.5 & 0.7 \end{bmatrix} + \begin{bmatrix} 0.5 & 0 \\ -0.5 & 0.3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus:

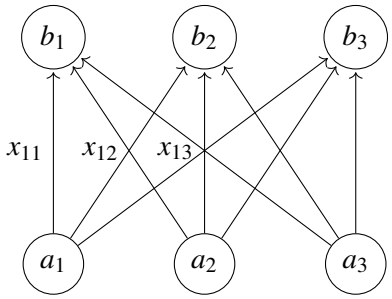
$$\mathbf{v}[n+1] = (\mathbf{A} + \mathbf{B}\mathbf{F})\mathbf{v}[n] = \mathbf{I}\mathbf{v}[n] = \mathbf{v}[n]$$

Since $\mathbf{v}[n+1] = \mathbf{v}[n]$ for all n, we can see that the measured velocity of the ball is constant!

6. Linear Neural Networks

Considered the pride and joy of modern-day machine learning, neural networks are simply the composition of basic mathematical functions that you already know about! In this problem, we will consider a very basic type of neural network, a *fully-connected linear neural network*, and derive closed-form solutions that describe the process used to train these networks to model complex behavior.

First, we will consider a single-layer neural network, in which a *linear combination* of the input values $a_1, a_2, \ldots, a_n$ are used to produce prediction values $b_1, b_2, \ldots, b_m$. An example of this architecture for $n = m = 3$ can be seen in the following figure:



In this case, we express mathematically that

$$b_j = \sum_{i=1}^{n} x_{ij} a_i$$

where each x_{ij} is the *weight* of the connection from input a_i to output b_j (note that only the weights from input a_1 are labeled in the diagram for visual reasons, but all input-output pairs do have appropriate weights). These model weights are the values that the model must "learn" from training examples in order to predict new outputs from future inputs.

(a) Using the provided mathematical expression, write b_j as the inner product between two vectors. Be sure to specify the entries of your vectors and what they represent.

Solution: The summation form $b_j = \sum_{i=1}^{n} x_{ij} a_i$ can be pattern-matched to the definition of the inner product between the vectors

$$\mathbf{x}_j = \begin{bmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Therefore, $b_j = \sum_{i=1}^{n} x_{ij} a_i = \mathbf{x}_j^\top \mathbf{a} = \langle \mathbf{x}_j, \mathbf{a} \rangle$.

Here, $\mathbf{a}$ represents all of our inputs to the neural network, while $\mathbf{x}_j$ represents the weights of each of the inputs to b_j.

(b) Let us define the vector $\mathbf{b} = \begin{bmatrix} b_1 & b_2 & \cdots & b_m \end{bmatrix}^\top$ (notice that this is a column vector). Using the inner product you derived above, express $\mathbf{b}$ as a matrix-vector multiplication. Be sure to specify the entries of your matrix and vector respectively, and describe what they represent. Note that this expression actually describes how we predict $\mathbf{b}$ given some inputs, a process known as *inference*.

Solution: We derive in the previous subpart that $b_j = \mathbf{x}_j^\top \mathbf{a}$. For vectors

$$\mathbf{x}_j = \begin{bmatrix} x_{1j} \\ x_{2j} \\ \vdots \\ x_{nj} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

In order to produce an expression for $\mathbf{b}$, we can "stack" our inner products (observing that $\mathbf{a}$ is the same for each b_j) to get

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & \cdots & x_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1m} & x_{2m} & \cdots & x_{nm} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

If we define

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

then $\mathbf{b} = \mathbf{X}^\top \mathbf{a}$.

As with the previous subpart, $\mathbf{a}$ represents all of our inputs to the neural network. The matrix $\mathbf{X}$ now represents all of our learned model weights x_{ij}.

If we actually want to use our neural network architecture, we first need to train it! In order to do this, we will estimate the weights x_{ij} in our network that best approximate the desired output $\mathbf{b}'$ given inputs $\{a'_1, a'_2, \ldots, a'_n\}$ (we can indicate these with the input vector $\mathbf{a}'$). It turns out that we'll need more than just one input-output pair to train our neural network, so let's define our dataset as the following pair of matrices:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mp} \end{bmatrix}$$

where $p \geq m, n$. For these matrices, a_{ki} is the value of a_i in the kth training input, and b_{jk} is the value of b_j in the kth training output. We can define each training pair $(\mathbf{a}_k, \mathbf{b}_k)$ as the kth **row** of matrix $\mathbf{A}$ and kth **column** of $\mathbf{B}$.

Assuming that our matrix-vector multiplication from the previous subpart can be written in the form $\mathbf{b} = \mathbf{X}^\top \mathbf{a}$, we want to find the matrix $\mathbf{X}$ that minimizes

$$\arg\min_{\mathbf{X} \in \mathbb{R}^{n \times m}} \|\mathbf{X}^\top \mathbf{A}^\top - \mathbf{B}\|_F^2$$

given our training data $(\mathbf{A}, \mathbf{B})$. The norm $\|\mathbf{M}\|_F$ of a matrix M is known as the *Frobenius norm* of that matrix.

As it turns out, this is actually the Least Squares problem, but with matrices instead! For this part, assume that the columns of $\mathbf{A}$ are linearly independent.

(c) Before we write an expression for the trained value of our entire matrix $\mathbf{X}$, let us consider a simpler version of the problem,

$$\arg\min_{\mathbf{x_j}\in\mathbb{R}^n} \|\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top\|^2$$

where $\mathbf{x}_j$ is the jth column of $\mathbf{X}$ and $\mathbf{b}_j^\top$ is the jth row of our output matrix $\mathbf{B}$ (you can verify this by matching up dimensions). Use Least Squares to find an expression for this optimal vector of weights, $\mathbf{x}_j^*$, in terms of $\mathbf{A}$ and $\mathbf{b}_j$.

Afterwards, convince yourself how this shows that $\mathbf{x}_j^*$ is the weight vector that best maps the values of all the inputs to the output b_j of our network.

Hint: What is the relationship between the norm of a vector and the norm of its transpose?

Solution: First, because the norm is equivalent between a vector and its transpose, we can rewrite the given optimization problem as

$$\arg\min_{\mathbf{x}_j\in\mathbb{R}^n} \|(\mathbf{x}_j^\top \mathbf{A}^\top - \mathbf{b}_j^\top)^\top\|^2$$

$$= \arg\min_{\mathbf{x}_j\in\mathbb{R}^n} \|\mathbf{A}\mathbf{x}_j - \mathbf{b}_j\|^2$$

By the Least Squares formula, we get that:

$$\mathbf{x}_j^* = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{b}_j$$

(d) Now that we have derived a closed-form solution for a particular set of weights $\mathbf{x}_j^*$, derive a closed-form solution for the optimal value of the weight matrix, $\mathbf{X}^*$, in terms of $\mathbf{A}$ and $\mathbf{B}$.

It may help to know that $\|\mathbf{M}\|_F^2 = \sum_{i=1}^{n} \|\mathbf{m}_i\|_2^2$, where $\{\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_n\}$ are the rows of $\mathbf{M}$. In other words, the squared Frobenius norm of a matrix is equal to the sum of the squared ℓ_2 norms of its rows.

Solution: We have derived that for each column of $\mathbf{X}$, the optimal assignment of weights is

$$\mathbf{x}_j^* = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{b}_j$$

Importantly, these optimal values are independent of the optimal values for the other columns of $\mathbf{X}$.

Per the description above regarding the Frobenius norm, we can rewrite our matrix optimization problem as

$$\arg\min_{\mathbf{X}\in\mathbb{R}^{n\times m}} \|\mathbf{X}^\top\mathbf{A}^\top - \mathbf{B}\|_F^2$$

$$= \arg\min_{\mathbf{X}\in\mathbb{R}^{n\times m}} \sum_{j=1}^{m} \|\mathbf{x}_j^\top\mathbf{A}^\top - \mathbf{b}_j^\top\|_2^2$$

since jth row of $\mathbf{X}^\top\mathbf{A}^\top - \mathbf{B}$ is $\mathbf{x}_j^\top\mathbf{A}^\top - \mathbf{b}_j^\top$. Because the optimal values of each $\mathbf{x}_j^*$ are independent of one another, we can additionally say that the optimal value of this reformulated optimization problem must simply be the $\mathbf{X}$ whose columns are the solutions to the individual Least Squares problem

$$\arg\min_{\mathbf{x}_j\in\mathbb{R}^n} \|\mathbf{x}_j^\top\mathbf{A}^\top - \mathbf{b}_j^\top\|^2 \qquad \forall j \in \{1, \ldots, m\}$$

Similar to part (b), we can observe that the matrix $(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ is consistent among the solutions for each column, so we can "stack" the closed-form solutions for each column $\mathbf{x}_j^*$ to get

$$\mathbf{X}^* = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{B}^\top$$

Note that we have $\mathbf{B}^\top$ instead of $\mathbf{B}$ since we are stacking the row vectors $\mathbf{b}_j^\top$ as columns. Using this solution, it turns out that we can directly compute the optimal weights for our neural network by simply computing an expression of our training inputs and outputs!

To learn more than you ever want to know about neural networks, take CS 189 and CS 182!

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7. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

(a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*

Solution:

I worked alone because my study group was out of town.

I prefer to work alone because a group can be very distracting for me.

I worked with my friends Alice and Bob. It's nice to be able to bounce ideas off one another and fill in each other's gaps. Explaining my ideas to someone else is hard, but it really makes sure I understand them well!

I met some new friends Carol and Danny at homework party, and we were able to finish most of the homework together with occasional help from a tutor. I reviewed my answers on my own to make sure I understood everything.

(b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

Solution:

I didn't remember some of the definitions I learned in linear algebra, so I consulted my old notes and reviewed concepts as I needed to while solving the problems

I went to office hours to work on the homework, and whenever I got stuck, I asked one of the tutors to provide some hints or review concepts I didn't remember.

I tried all the problems on my own, and then asked an LLM to critique my approach. I got 90% of the way there on my own, but this interaction enriched my understanding of the material!

I reviewed my lecture notes before starting the homework; and whenever I got stuck, I scoured the discussion worksheets for similar problems and used the solutions to guide my thought process.

I asked some questions on Ed whenever I was confused so that I could get the most fitting answer to my question, in the context of EE66. I also answered some questions because this is a good way to practice the stuff I know well.

I couldn't even understand what problem 2 was asking, so I asked an AI tool to review all the relevant prerequisite knowledge and explain the problem statement in a different way before trying to tackle it on my own.

(c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

Solution:

I found question 3 very intriguing because this is a totally different perspective on DTFS to what we learned in lecture. Question 2 was pretty boring and mechanical, because we were just practicing the synthesis equation, but I suppose it is important to be able to do these more-basic things too.

(d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

Solution: I did feel prepared! But, question 5 seemed to me to be out of place because it's been two weeks since we covered least squares. We talked about SVD and its application for PCA in class and discussion this week, and I didn't really understand any of it, so it would have been nice to have some problems which gave me an opportunity to practice those concepts.

(e) Is there anything else you would like to share?

Solution:

Johnny, who is a tutor, was SUPER DUPER helpful in office hours. He explained stuff in such a way that I still felt like I was discovering the answer to the problems by myself, which was awesome.

I'm feeling very overwhelmed by this class and the upcoming midterm is really stressing me out. These homeworks are really hard and I wish there were more resources available, which I could use to understand them deeply.

Temporary page!

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